

Beyond E2E Scripts: Using LLM-Proposed Scenarios Without Letting the LLM Be the Oracle

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Abstract

Large language models can help testers explore application states that are not covered by ordinary end-to-end scripts, but they are unsafe as test oracles. They can suggest plausible scenarios, misread application semantics, and overstate weak evidence. Bayesilisk separates these roles. App-specific connectors publish observable context, available actions, and deterministic rules. Bayesilisk expands the supplied scenario space, ranks probes, and verifies only observed evidence returned by the connector. This draft reports an initial Cal.com case study at commit `180ede28f0bddf2738933a6e60a8e80f6116d7da`: connector-supplied route facts and an action graph produced seven Bayesilisk probe proposals. Six local Playwright executions reproduced a mismatch where unknown or stale `rescheduleUid` values opened booking flows with semantic status 200 although the supplied invariant expected 404; a seventh bounded workflow proposal created a booking, cancelled it, and replayed the cancelled booking UID through the public booking route, again observing 200 where the supplied invariant expected 409. One reported finding already has an upstream human-authored fix pull request. The current artifact set keeps only this clean seven-finding rerun; baseline comparisons should be recreated as fresh runs when needed.

1 Problem

E2E suites are precise but narrow. LLM-driven agents are broad but unreliable: they can drive a browser, invent scenarios, and then incorrectly decide whether the behavior is acceptable. The core research claim behind Bayesilisk is that these jobs should be separated. An LLM or coding agent may help propose scenario spaces, but deterministic verifier rules must decide pass or fail.

The intended workflow is:

1. A connector extracts app-specific context: routes, actors, identifiers, available actions, nearby tests, and expected behavior.
2. Bayesilisk expands only the supplied rule space and produces probe proposals.
3. The connector executes those proposals through the real app or its test harness.
4. Bayesilisk verifies returned evidence against deterministic invariants and emits issue-ready findings.

The connector is app-specific. The Bayesilisk core is not. This boundary is the main architectural point: app teams write connectors; Bayesilisk provides the generic expansion, ranking, and verification loop.

2 Cal.com Case Study

We evaluated the current connector architecture on a local Cal.com checkout. The tested repository and revision were:

- Repository: <https://github.com/calcom/cal.com>
- Commit: `180ede28f0bddf2738933a6e60a8e80f6116d7da`

- Commit date: 2026-05-14 19:30:21 +0000
- Commit subject: font-stack fallback fix for non-Latin script support (#29346)
- Bayesilisk seed: 150

The checked-in Bayesilisk artifacts are in `examples/calcom/`. They include the connector, source contexts, generated proposals, consolidated execution context, app-only JSON report, app-only Markdown report, issue payloads, and the two Playwright run logs for the clean rerun.

Artifact	Path
Connector code	<code>examples/calcom/bayesilisk-probes.e2e.ts</code>
Source context	<code>examples/calcom/source-context.json</code>
Generated proposals	<code>examples/calcom/generated-proposals.json</code>
Sequence context	<code>examples/calcom/sequence-source-context.json</code>
Generated sequence	<code>examples/calcom/generated-sequence-proposals.json</code>
Execution evidence	<code>examples/calcom/execution-context.json</code>
Bayesilisk report	<code>examples/calcom/reports/report.json</code>
Issue payloads	<code>examples/calcom/reports/issue-payloads.json</code>
Upstream outcomes	<code>examples/calcom/upstream-outcomes.md</code>

Table 1: Checked-in Cal.com artifacts.

The connector supplied source facts for Cal.com routes and an action graph for one workflow. Bayesilisk generated six route probes and one workflow-sequence probe. The route probes mutate unknown and stale `rescheduleUid` values for public, private, and dynamic booking routes. The workflow probe creates a booking, cancels it, and replays the cancelled booking UID through the public booking route. The connector executed all seven proposals with Playwright against local fixtures.

Area	Mutation	Expected	Observed
Public booking page	unknown id	404	200
Public booking page	stale id	404	200
Private booking link	unknown id	404	200
Private booking link	stale id	404	200
Dynamic booking page	unknown id	404	200
Dynamic booking page	stale id	404	200
Cancelled booking replay	cancelled uid	409	200

Table 2: Verified Cal.com connector observations.

The result is useful because the pass/fail verdict did not come from a model. The connector returned concrete observed facts. Bayesilisk compared `expectedStatus` and `observedStatus`, classified the seven observations as `breakage.context-observed`, and produced issue payloads. Two upstream issues were opened from this evidence: one for public `rescheduleUid` replay [1] and one for cancelled booking UID replay [2]. The first issue already has a human-authored fix pull request [3]. This is stronger validation than issue closure alone: an issue can be closed as erroneous, but a targeted fix PR indicates that a human contributor found the report actionable.

3 Upstream Response

The upstream response is important because the claim is not only that Bayesilisk can emit plausible reports. The reports need to be useful to humans working on the target system.

The private-link finding is tracked as a related comment rather than a separate issue to avoid duplicating the existing `rescheduleUid` thread. The cancelled-booking workflow finding is separate because it comes

Finding group	Issue	Human response	Status at capture
Public unknown/stale <code>rescheduleUid</code>	#29399	Fix PR #29400	Open
Private-link <code>rescheduleUid</code>	#29399 comment	Folded into thread	Open
Cancelled booking UID replay	#29407	Newly reported	Open

Table 3: Upstream response to Bayesilisk-generated Cal.com findings.

from a different state transition: a real booking is created, cancelled, and then replayed as public reschedule context.

4 Evidence Still Needed

The Cal.com result demonstrates the connector split and a real route/state mutation finding. It does not yet demonstrate the full paper claim. The remaining evidence needs are fresh baseline comparisons and additional app artifacts beyond Cal.com.

Workflow	Observed scope	Failures	Notes
Bayesilisk proposal run	6 proposals	6	Unknown and stale variants
Bayesilisk sequence run	1 sequence	1	Cancelled UID replay

Table 4: Current clean Cal.com rerun.

Fresh baselines matter because the argument is not merely that browser automation can find a failing route. The intended claim is stronger: generic context-to-probe expansion plus deterministic verification can recover useful failures while avoiding the oracle mistakes of plain LLM agents.

5 Discussion

The current evidence supports a narrow claim: given connector-supplied source facts and rules, Bayesilisk can expand a small scenario space, execute it through an app-specific connector, and verify evidence without letting the LLM decide the outcome. This is already different from a plain Playwright script because the connector does not enumerate every final generated probe by hand; it exposes the local state and rule space, and Bayesilisk produces the concrete mutations.

The remaining work is empirical. The paper needs more app artifacts and architecture work before it can claim generality for longer workflows. The new sequence proposer is intentionally bounded: connectors declare actions and state facts, while Bayesilisk generates short action sequences and verifies returned evidence. The immediate target should be a concise artifact-backed paper rather than a broad methodology claim.

6 Conclusion

Bayesilisk treats LLMs as scenario proposers, not as test oracles. The Cal.com case study shows the architecture working on a real app checkout with reproducible connector evidence and issue-ready findings. The next version of this manuscript should add a full second app artifact and recreate baseline comparisons against existing E2E tests, Playwright-only probing, weak model proposals, and browser-driving LLM agents.

References

- [1] Bayesilisk contributors. Cal.com issue 29399: Booking page opens instead of 404 for missing rescheduleuid. <https://github.com/calcom/cal.diy/issues/29399>, 2026. Issue report generated from Bayesilisk evidence.
- [2] Bayesilisk contributors. Cal.com issue 29407: Cancelled booking uid replay opens public booking route. <https://github.com/calcom/cal.diy/issues/29407>, 2026. Issue report generated from Bayesilisk workflow-sequence evidence.
- [3] Cal.com contributor. Cal.com pull request 29400: Return 404 for unknown rescheduleuid instead of opening booking flow. <https://github.com/calcom/cal.diy/pull/29400>, 2026. Human-authored fix pull request responding to Bayesilisk evidence.